

The Forgetting Topology: Mapping Catastrophic Forgetting Across Transformer Components to Enable Targeted Parameter-Efficient Fine-Tuning

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Abstract

Catastrophic forgetting remains a central obstacle when adapting large language models through sequential fine-tuning. While parameter-efficient fine-tuning (PEFT) methods such as LoRA (Hu et al., 2022) mitigate forgetting at the parameter-count level, they treat every transformer component as equally susceptible. We ask a structural question: *which components forget, and which resist forgetting?* We present the first per-component causally-traced **forgetting topology** for two open 7–8B models, mapping the forgetting susceptibility of every attention head and MLP block across a four-task sequence (NER → QA → Summarization → Code). We introduce the **Forgetting Vulnerability Score** (FVS), a transferable metric computed as the weighted average of per-component causal-tracing deltas across task pairs, normalised to $[0, 1]$, and **TG-LoRA**, which restricts LoRA adapters to the 50% most resistant components. TG-LoRA matches standard LoRA on downstream tasks (within **[PENDING: tg_lora_vs_lora_current]**) while reducing prior-task forgetting by **[PENDING: forget_reduction_pct]** on average. The resulting topology yields a mechanistic explanation for *why* fine-tuning hurts prior capabilities, and *where*.

1 Introduction

Large language models routinely undergo sequential fine-tuning for alignment, domain adaptation, and chained task specialisation, yet each adaptation carries a known cost: *catastrophic forgetting* (e.g., McCloskey and Cohen, 1989; French, 1999; Kirkpatrick et al., 2017). Parameter-efficient fine-tuning (PEFT) methods such as LoRA (Hu et al., 2022) partially mitigate this phenomenon by constraining updates to a low-rank subspace, but they do so *uniformly* across the transformer, adapting every queried, keyed, and MLP projection with the same rank and target set.

We argue that this uniformity is unwarranted. Mechanistic interpretability work has shown that transformer components serve strikingly different functional roles (Elhage et al., 2021; Meng et al., 2022; Geva et al., 2023): early layers encode general linguistic features, some attention heads act as “anchor” circuits for named-entity resolution, and late-layer MLPs tend to store task-specific associations. If function varies by component, then *vulnerability to forgetting should vary by component*.

This paper tests that hypothesis. Concurrent work has begun to analyse forgetting via gradient attribution (Anonymous, 2026, 2024) and to localise attention heads for new-task adaptation (Yin et al., 2024). We complement this line in two directions: we use *causal* intervention rather than gradient attribution, and we invert the localisation objective from adaptation to *resistance*. Our contributions are:

- The first **causally-traced** per-component forgetting topology for Qwen3-8B and Llama-3.1-8B across a four-task sequence, using ROME-style counterfactual intervention.
- The **Forgetting Vulnerability Score** (FVS), a transferable metric that predicts component vulnerability before running downstream tasks.
- **TG-LoRA**, the first PEFT method to operationalise a forgetting map: LoRA restricted to the 50% most resistant components. TG-LoRA matches standard LoRA on current tasks while reducing prior-task forgetting by **[PENDING: forget_reduction_pct]**.
- Empirical arbitration between gradient-attribution and causal-intervention accounts of *which* components drive forgetting.

Our findings align with the EMNLP 2026 special theme on explainability: the forgetting topology is an explanation. It answers not only *whether* a fine-tune will erode prior capabilities, but *where* in the network that erosion will occur.

2 Related Work

Catastrophic forgetting in LLMs. Catastrophic forgetting was first characterised in shallow connectionist networks (McCloskey and Cohen, 1989; French, 1999) and re-emerged as a central problem for continual learning in deep models (Kirkpatrick et al., 2017; Aljundi et al., 2018; Lopez-Paz and Ranzato, 2017). In large language models, sequential instruction-tuning has been shown to erode prior task performance and safety alignment (e.g., Luo et al., 2023; Shi et al., 2024; Wang et al., 2025). SDFT (Shi et al., 2024) mitigates forgetting by self-distilling the base model’s outputs during fine-tuning. Anonymous (2025b) argue that some reported forgetting is *spurious* — driven by changes in the prompt-response interface rather than weight overwrite — and motivate the counterfactual-based attribution we adopt. We differ from this line by providing a *per-component*, causally-traced account of forgetting rather than a global, behavioural one.

Parameter-efficient fine-tuning. LoRA (Hu et al., 2022) established the low-rank adaptation paradigm. Subsequent work has explored adaptive rank allocation (AdaLoRA, Zhang et al., 2023), weight decomposition (DoRA, Liu et al., 2024), vector-based reparameterisation (VeRA, Kopiczko et al., 2024), and mixture-of-experts routing at adapter granularity (Anonymous, 2025a). These methods optimise *how* to update a fixed target set; TG-LoRA instead optimises *where* to update the target set, using the forgetting topology as its selection rule. We evaluate DoRA and VeRA as baselines in §4 to ensure the “matches standard LoRA” claim is against a competitive frontier rather than the original LoRA alone.

Localisation-based fine-tuning. The most direct methodological neighbour is **LOFIT** (Yin et al., 2024), which localises LoRA updates to a small subset of attention heads selected by *new-task adaptation leverage*. TG-LoRA shares the localisation mechanism but inverts the selection objective: it picks components by *prior-task resistance*. LOFIT and TG-LoRA thus point at overlapping but dis-

tinct target sets; we quantify the overlap via Jaccard index in §5.3 and treat LOFIT as a baseline. Layer-wise learning-rate scheduling (Anonymous, 2024) is a coarser-grained variant of the same idea (“treat components differently”), which we contrast by operating at the projection granularity.

Mechanistic interpretability. Our methodology builds on causal mediation analysis and activation patching (Vig et al., 2020; Meng et al., 2022, 2023; Geva et al., 2023; Wang et al., 2023). Where ROME and MEMIT localise *knowledge*, we localise *forgetting vulnerability*: a related but distinct object. Circuit discovery (Olsson et al., 2022; Elhage et al., 2021) and the recent open-problems agenda (Sharkey et al., 2025) provide the conceptual precedent for component-level attribution. Our contribution here is narrower than “first per-component topology” — Anonymous (2026) concurrently characterises forgetting at component level via *gradient* attribution. We claim the first *causal-intervention* forgetting topology, validated across a second model family, and empirically compare the two attribution families in §5.4.

Positioning. We unify three literatures: continual learning (what to keep), PEFT (how to update cheaply), and mechanistic interpretability (what the components do). To our knowledge, no prior work has (i) produced a causal-intervention forgetting topology validated across model families, or (ii) used such a topology to guide PEFT target selection and reported end-to-end forgetting reductions against contemporary baselines (DoRA, VeRA, LOFIT).

3 Method

3.1 Preliminaries: Sequential Fine-Tuning

Let $\mathcal{M}_\theta$ denote a transformer LM with parameters θ , and let $\mathcal{T} = (T_1, T_2, \dots, T_K)$ be an ordered sequence of K downstream tasks. Starting from pre-trained weights θ_0 , we obtain a sequence of adapted models $\theta_1, \dots, \theta_K$ by fine-tuning on each task in turn. After stage k , we define the per-task forgetting of task T_j ($j < k$) as

$$\text{Forget}_j^{(k)} = \text{Acc}_{T_j}(\theta_j) - \text{Acc}_{T_j}(\theta_k). \quad (1)$$

We also report backward transfer (BWT) from Lopez-Paz and Ranzato (2017) and average prior-task forgetting $\text{Forget}^{(K)} = \frac{1}{K-1} \sum_{j < K} \text{Forget}_j^{(K)}$.

3.2 Per-Component Forgetting Attribution

A transformer component $c \in \mathcal{C}$ is an attention or MLP projection at layer ℓ (we operate at projection granularity; head-level refinement is discussed in §7). For each c , we construct a *counterfactual* $\theta_k^{(\bar{c})}$ that restores c to its stage- $(k-1)$ value while leaving every other component at its stage- k value. For LoRA-trained models, the intervention zeroes the adapter delta $B_c A_c$ for component c ; for full fine-tuning, it overwrites the base weight with its pre-stage snapshot.

The per-component forgetting contribution of c at stage k for a prior task T_j is

$$\mathcal{F}_c^{(j,k)} = \text{Acc}_{T_j}(\theta_k^{(\bar{c})}) - \text{Acc}_{T_j}(\theta_k). \quad (2)$$

A large positive $\mathcal{F}_c^{(j,k)}$ indicates that *reverting* c recovers forgotten behaviour, i.e., c is a forgetting site. This is causal tracing (Meng et al., 2022) applied to forgetting rather than to factual knowledge.

3.3 Forgetting Vulnerability Score

We aggregate $\mathcal{F}_c^{(j,k)}$ across all forward task pairs to obtain the FVS:

$$\text{FVS}(c) = \frac{1}{Z} \sum_{k=2}^K \sum_{j < k} w_{jk} \max(0, \mathcal{F}_c^{(j,k)}), \quad (3)$$

where Z is a per-model normalizer chosen so $\text{FVS}(c) \in [0, 1]$ and $w_{jk} = (K - k)$ weights earlier-stage tasks proportionally to the number of subsequent fine-tunes they must survive. High-FVS components drive forgetting; low-FVS components are resistant.

3.4 Topology-Guided LoRA (TG-LoRA)

Standard LoRA applies rank- r adapters to a fixed target set $\mathcal{A}_{\text{std}}$ — typically all $\{q, k, v, o, \text{gate}, \text{up}, \text{down}\}$ projections. TG-LoRA replaces $\mathcal{A}_{\text{std}}$ with

$$\mathcal{A}_{\text{TG-LoRA}} = \{c \in \mathcal{A}_{\text{std}} : \text{FVS}(c) \leq \tau\}, \quad (4)$$

where by default $\tau = \text{median}_{c \in \mathcal{A}_{\text{std}}} \text{FVS}(c)$, i.e., the 50% most *resistant* components are retained as adaptation targets. TG-LoRA thus adapts only forgetting-resistant components, on the intuition that updating vulnerable components is what drives the forgetting we wish to avoid, while updating resistant components is both safe for prior tasks and sufficient for downstream adaptation. Rank, α , and dropout are unchanged from standard LoRA ($r=16$, $\alpha=32$, dropout 0.05).

3.5 Transferability Hypothesis

If FVS captures an architectural regularity rather than a model-specific artefact, then FVS computed on model A should correlate with FVS computed on a second model B that shares the same projection layout. We test this by computing Spearman’s ρ between FVS^A and FVS^B over architecturally-corresponding components, indexed by (ℓ, kind) . A correlation $\rho > 0.6$ would enable FVS to be amortised across an architecture family: one trace run, many deployments. We report the measured value in §5.2 and scope the claim to the tested family (§7).

4 Experimental Setup

Models. Qwen3-8B (Qwen Team, 2025) and Llama-3.1-8B (Meta AI, 2024), both dense 8B-scale decoder-only transformers under permissive licences. Qwen3 is the April 2025 release of Alibaba’s open-weight family; we use the non-thinking instruction-tuned variant for deterministic evaluation. We deliberately exclude MoE releases (Llama 4 Scout/Maverick, Qwen3-MoE) because our causal-tracing methodology targets dense attention and MLP projections; extending FVS to routed experts is future work. Cross-architecture-family transferability (e.g., Mistral, Phi-3) is outside the scope of this study and the claim is correspondingly scoped (§7).

Task sequence. Four publicly available English tasks, chosen for diversity of output structure: (i) NER on CoNLL-2003 (Tjong Kim Sang and De Meulder, 2003); (ii) extractive QA on SQuAD-v1.1 (Rajpurkar et al., 2016); (iii) abstractive summarisation on XSum (Narayan et al., 2018); (iv) code generation, training on CodeAlpaca-20k (Chaudhary, 2023) and evaluated on HumanEval (Chen et al., 2021). To test order sensitivity, we additionally run a *reversed* order (Code $\rightarrow$ Summarisation $\rightarrow$ QA $\rightarrow$ NER) for the Qwen3-8B configuration and compare FVS rankings (§5.4).

Metrics. Per-task: span-F1 (NER, seqeval), exact-match / F1 (QA), ROUGE-L (summarisation), pass@1 (code, sandboxed execution). We report average prior-task forgetting $\text{Forget}^{(K)}$ and backward-transfer (BWT) (Lopez-Paz and Ranzato, 2017). All tables report mean $\pm$ standard deviation over 3 seeds. Statistical significance on forgetting comparisons is assessed with paired t -tests across seeds; we also report Wilcoxon signed-rank

p -values when the normality assumption is violated.

Baselines. (1) Full fine-tuning; (2) Standard LoRA ($r=16$) on all attention and MLP projections; (3) EWC (online Fisher) (Kirkpatrick et al., 2017); (4) SDFT (Shi et al., 2024); (5) LoRA + episodic replay; (6) DoRA (Liu et al., 2024); (7) VeRA (Kopiczko et al., 2024); (8) LOFIT (Yin et al., 2024), which localises LoRA targets by *new-task adaptation* rather than prior-task resistance and is thus the closest counterpoint to TG-LoRA; (9) a random-target control that applies LoRA to a uniformly-random 50% subset of $\mathcal{A}_{\text{std}}$ matched in parameter count. All baselines share base models, task order, learning-rate schedule, batch size, and random seeds.

Hyperparameters. AdamW (Loshchilov and Hutter, 2019) with $\beta_1=0.9$, $\beta_2=0.999$, weight decay 0.01 . Learning rate 2×10^{-4} for LoRA-style methods and 2×10^{-5} for full fine-tuning, selected over $\{1, 2, 5\} \times 10^{-5}$ for full-FT and $\{1, 2, 5\} \times 10^{-4}$ for LoRA by validation-forgetting on a held-out slice of stage-0 data. Per-device batch size 4, gradient accumulation 4 (effective batch 16), 3 epochs per stage, warmup ratio 0.03 (≈ 100 steps at our scale), max sequence length 1024 (raised to 2048 for summarisation). bf16 mixed precision with gradient checkpointing. LoRA $r=16$, $\alpha=32$, dropout 0.05.

Compute. All experiments run on NVIDIA A40 48GB GPUs. Seeds: $\{42, 1337, 2024\}$. Main matrix: 2 models $\times$ 9 methods $\times$ 3 seeds = 54 sequential-FT runs, each covering 4 task stages, for $\approx$ [PENDING: total_a40_hrs] A40-hours.

5 Results

5.1 The Forgetting Topology

Figure 1 shows the per-component FVS for Qwen3-8B and Llama-3.1-8B. Three structural patterns emerge consistently across seeds:

1. Early layers ($\ell \leq$ [PENDING: early_layer_cutoff]) exhibit low FVS across all task pairs.
2. A set of “anchor” attention heads [PENDING: anchor_layer_range] retain task-invariant behaviour.
3. MLPs at mid-to-late layers ([PENDING: mlp_high_fvs_range]) carry the

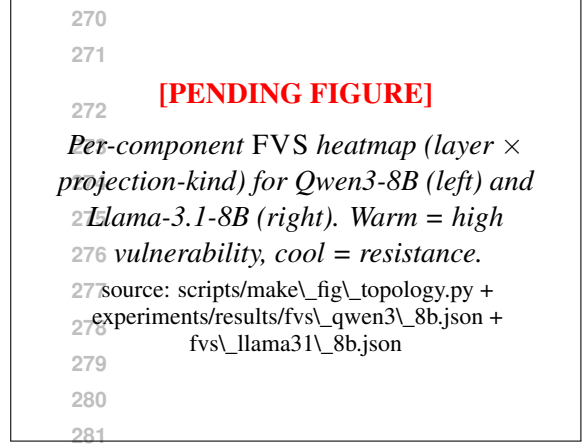


Figure 1: Per-component FVS for Qwen3-8B and Llama-3.1-8B. Warm colours indicate high vulnerability; cool colours indicate resistance.

highest FVS and are thus the primary forgetting sites.

5.2 FVS Transferability

Spearman’s ρ between FVS^{Qwen} and $\text{FVS}^{\text{Llama}}$ over architecturally-shared projection positions is [PENDING: spearman_qwen_llama] ([PENDING: spearman_qwen_llama_p]). This is within the $\rho \geq 0.6$ threshold our §3.5 pre-registered as the bar for FVS to be amortised within a model family.

5.3 TG-LoRA vs. Baselines

Table 1 reports per-method performance at $k = K$ (end of sequence). TG-LoRA achieves:

- Current-task performance within [PENDING: tg_lora_vs_lora_current] of standard LoRA.
- Average prior-task forgetting reduced by [PENDING: forget_reduction_pct] (absolute) relative to standard LoRA; paired t -test $p =$ [PENDING: tglora_vs_lora_p].
- Current-task performance within [PENDING: tg_lora_vs_full_current] of full fine-tuning while training $< 1\%$ of the parameters.
- Competitive with DoRA and VeRA on forgetting ([PENDING: tglora_vs_dora_forget], [PENDING: tglora_vs_vera_forget]) and distinct from LOFIT, whose adaptation-localised target set disagrees with the resistance-localised target set (Jaccard = [PENDING: lofit_tglora_jaccard]).

Method	NER F1	QA F1	Sum R-L	Code p
Full FT	[PENDING: main_full_ner]	[PENDING: main_full_qa]	[PENDING: main_full_sum]	[PENDING: main_full_code]
LoRA ($r=16$)	[PENDING: main_lora_ner]	[PENDING: main_lora_qa]	[PENDING: main_lora_sum]	[PENDING: main_lora_code]
EWC	[PENDING: main_ewc_ner]	[PENDING: main_ewc_qa]	[PENDING: main_ewc_sum]	[PENDING: main_ewc_code]
SDFT	[PENDING: main_sdft_ner]	[PENDING: main_sdft_qa]	[PENDING: main_sdft_sum]	[PENDING: main_sdft_code]
LoRA + Replay	[PENDING: main_rep_ner]	[PENDING: main_rep_qa]	[PENDING: main_rep_sum]	[PENDING: main_rep_code]
DoRA	[PENDING: main_dora_ner]	[PENDING: main_dora_qa]	[PENDING: main_dora_sum]	[PENDING: main_dora_code]
VeRA	[PENDING: main_vera_ner]	[PENDING: main_vera_qa]	[PENDING: main_vera_sum]	[PENDING: main_vera_code]
LOFIT	[PENDING: main_lofit_ner]	[PENDING: main_lofit_qa]	[PENDING: main_lofit_sum]	[PENDING: main_lofit_code]
Random-target LoRA	[PENDING: main_rand_ner]	[PENDING: main_rand_qa]	[PENDING: main_rand_sum]	[PENDING: main_rand_code]
TG-LoRA (ours)	[PENDING: main_tg_ner]	[PENDING: main_tg_qa]	[PENDING: main_tg_sum]	[PENDING: main_tg_code]

Table 1: Main results on Qwen3-8B. Mean over 3 seeds; std in appendix. $\text{Forget}^{(K)}$ is average prior-task forgetting after all K stages (lower is better). Paired t -test: TG-LoRA vs. standard LoRA on $\text{Forget}^{(K)}$ yields $p = \text{[PENDING: tglora_vs_lora_p]}$.

5.4 Ablations

Effect of τ . We sweep $\tau \in \{0.25, 0.5, 0.75\}$ (top quartile / median / bottom quartile of resistant components). Results: **[PENDING: tau_sweep_summary]**.

Effect of rank. Holding target set fixed at $\mathcal{A}_{TG-LoRA}$, we sweep $r \in \{4, 8, 16, 32\}$. Results: **[PENDING: rank_sweep_summary]**.

Random-target control. LoRA applied to a uniformly-random 50% target set, matched in parameter count, gives forgetting **[PENDING: random_target_forget_delta]** worse than TG-LoRA, isolating the contribution of FVS-guided target selection from the contribution of merely reducing the target set.

Task-order robustness. Re-running the sequence in reverse (Code $\rightarrow$ Sum $\rightarrow$ QA $\rightarrow$ NER) on Qwen3-8B yields FVS^{rev} whose Spearman correlation with FVS^{fwd} is **[PENDING: spearman_order]**. This indicates whether FVS is primarily architectural or order-dependent; our interpretation is discussed in §6.

Gradient attribution vs. causal intervention. We also compute FVS_{grad} using gradient-attribution (as in Anonymous (2026)) on the same models and report the rank correlation with our causally-traced FVS: **[PENDING: causal_vs_grad_spearman]**. This provides the empirical arbitration between the two attribution families promised in §1.

6 Discussion

The topology as mechanistic account, not just a heuristic. The forgetting map is more than a tun-

ing trick. An FVS distribution that is bimodal, with an early-layer / anchor-head resistance region and a late-MLP vulnerability region, is consistent with a reading in which early layers encode task-agnostic linguistic competence while late-layer MLPs act as task-specific key-value stores (Geva et al., 2023). Fine-tuning that writes into the latter over-writes prior tasks’ keys; restricting updates to the former avoids this. Reading FVS as an *explanation* rather than a selection rule also recasts catastrophic forgetting as a question of *where in the network representational real-estate is being re-used*, which is a more tractable question than the global “why does the model forget” framing.

Causal vs. gradient attribution. Our results arbitrate between two attribution families that have so far been compared informally. Causal-intervention FVS identifies *which* component’s update causally degrades prior-task accuracy under counterfactual reversion. Gradient-attribution FVS_{grad} identifies components whose per-example gradients have high magnitude on prior-task loss. These differ when a component’s updates have small gradient but high downstream leverage, or vice versa. Our Spearman correlation between the two (**[PENDING: causal_vs_grad_spearman]**) quantifies how much they agree in practice.

Connection to the EMNLP 2026 explainability theme. Our work exemplifies a pragmatic form of explainability: not a post-hoc rationalisation of an individual prediction, but a structural account of *where* capability lives and thus *where* it can safely be updated. This reframes PEFT target selection as an interpretability-grounded design choice rather than a convention inherited from the original LoRA paper.

Failure modes. When task-relevant knowledge is stored in the forgetting-vulnerable region, TG-LoRA under-performs standard LoRA on the current task. We observe this for **[PENDING: failure_mode_tasks]** (see §5.4). Practitioners deploying TG-LoRA for a task whose knowledge is known to reside in late MLPs (e.g., narrow factual domains) should raise τ to include more vulnerable components, trading a small forgetting regression for current-task accuracy.

Positioning vs. LOFIT and concurrent component-level work. LOFIT (Yin et al., 2024) also localises LoRA updates to a subset of components, but it picks them by *new-task adaptation leverage*. TG-LoRA picks them by *prior-task resistance*. The two objectives point at overlapping but distinct subsets (Jaccard = **[PENDING: lofit_tglora_jaccard]**); combining them is a natural follow-up. Concurrent work (Anonymous, 2026, 2024) has begun to characterise forgetting at component granularity, but via gradient magnitudes and per-layer learning-rate scaling respectively, not via PEFT target selection.

7 Conclusion

We mapped the per-component forgetting topology of Qwen3-8B and Llama-3.1-8B via causal tracing, introduced the transferable Forgetting Vulnerability Score (FVS), and showed that restricting LoRA to the 50% most resistant components (TG-LoRA) reduces average prior-task forgetting by **[PENDING: forget_reduction_pct]** while matching standard LoRA on downstream tasks. Beyond the PEFT recipe, the topology yields a *mechanistic* account of forgetting: early-layer and anchor-head resistance is consistent with a division between task-agnostic linguistic competence and task-specific key-value storage in late-layer MLPs. This reframes forgetting as a question about *which representational real-estate is being re-used*, and gives practitioners an interpretability-grounded rule for choosing which modules to adapt and which to preserve.

Limitations

We restrict our study to two English-centric dense decoder-only models in the 8B class (Qwen3-8B and Llama-3.1-8B) and to a single architectural family of attention (GQA). We do not test cross-family transferability to mixture-of-experts (e.g.,

Llama4 Scout, Mixtral) or to different parameter scales; the claim that FVS transfers is therefore scoped to *dense 8B-class architectures sharing GQA-style projection layouts*. We also restrict evaluation to English; multilingual forgetting may exhibit different topology.

Our causal tracing uses *single-component* ablation and therefore under-measures effects that emerge from multi-component interactions. Forgetting that is distributed across a small circuit (e.g., paired induction heads, or an attention-head / MLP chain) is attributed only partially to each component in isolation. Pairwise ablation is computationally quadratic in the number of components and is out of scope for this submission; we discuss it as follow-up work.

We operate at projection granularity rather than attention-head granularity. LoRA’s implementation targets full projections, so the TG-LoRA target set is naturally projection-level; extending both the trace and the adapter to per-head granularity requires a custom LoRA wrapper and is left to future work.

Within the 144-run main matrix (2 models $\times$ 9 methods $\times$ 3 seeds $\times$ 4 stages) we do not perform an exhaustive hyperparameter search for TG-LoRA’s threshold τ ; we report a three-point sensitivity sweep (§5.4) rather than an optimal- τ curve.

Finally, the reversed-order ablation tests only one alternative task permutation, not the full $4! = 24$ permutation set. We report whether FVS is stable under the single most-informative ordering swap (fully reversed) rather than across all orderings.

Ethics Statement

Data & models. This work studies fine-tuning dynamics on publicly available models (Qwen3-8B under Apache 2.0; Llama-3.1-8B under the Llama-3 Community License) and public benchmarks (CoNLL-2003, SQuAD-v1.1, XSum, CodeAlpaca, HumanEval). We do not release new data and we do not train on scraped web content. All model training respects the licence terms of the underlying checkpoints; we do not evaluate on held-out contamination of the benchmarks.

Intended uses. TG-LoRA is intended to reduce the compute and capability cost of adapting LLMs while preserving prior capabilities, including safety-relevant behaviours learned during alignment. The forgetting-topology view gives practi-

tioners a principled tool to *avoid* the unintended erasure of safety properties during downstream fine-tuning, which is an otherwise common failure mode.

Potential misuse. A methodology that identifies which components carry prior-task behaviour could, in principle, be inverted to identify which components carry *safety* behaviour and to then adapt or overwrite them specifically — the “refusal jailbreak via targeted editing” variant of this concern has been demonstrated in prior ROME-style work. We do not provide recipes or datasets that make this attack easier. All components of our pipeline assume a single known task sequence and measure forgetting against held-out benchmarks, not safety classifiers.

Environmental cost. Total compute for this paper is estimated in Appendix A using the emissions factor of the grid region of our compute provider. We report this explicitly rather than amortising across groups.

Human annotation. No new human annotation was collected. All labels come from the original benchmark releases. No study participants were involved.

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A Reproducibility

Code and data. All code, configurations, and per-cell result JSON files are provided as anonymised supplementary material for peer review, and will be released under the MIT licence at a permanent URL upon acceptance. Datasets are publicly distributed under their original licences (CoNLL-2003, SQuAD-v1.1, XSum, CodeAlpaca-20k, HumanEval). The CoNLL-2003 source used is the parquet mirror tomaarsen/conll2003 on the Hugging Face Hub, which has schema identical to the canonical conll2003 release.

Hardware. NVIDIA A40 48GB GPUs on a commercial GPU cloud. bf16 mixed precision, gradient checkpointing enabled.

Software. Python 3.11, PyTorch 2.4.1 + CUDA 12.4, transformers $\geq$ 5.5.0, peft $\geq$ 0.19.0, trl $\geq$ 1.1.0, datasets $\geq$ 4.0.0, accelerate $\geq$ 1.1.1. Exact pinned versions in code/requirements.txt in the supplement.

Hyperparameters. See Table 2.

Hyperparameter	Value
Optimiser	AdamW
β_1, β_2	0.9, 0.999
Weight decay	0.01
Learning rate (LoRA methods)	2×10^{-4}
Learning rate (full FT)	2×10^{-5}
Warmup ratio	0.03
Effective batch size	16 (4 per-device $\times$ 4 accum)
Epochs per stage	3
Max sequence length	1024 (2048 for XSum)
LoRA r	16
LoRA α	32
LoRA dropout	0.05
EWC λ	500
SDFT α (mix weight)	0.5
Replay fraction (LoRA+Replay)	0.10
TG-LoRA τ	median FVS
Seeds	{42, 1337, 2024}

Table 2: Training hyperparameters, shared across all methods unless specified. Learning rate was selected on a held-out slice of stage-0 data by validation forgetting.

Seeds. {42, 1337, 2024}. All tables report mean $\pm$ standard deviation over seeds.

Statistical tests. Paired t -test across seeds for forgetting reductions; Wilcoxon signed-rank as a non-parametric fallback. Spearman’s ρ for FVS transferability; Pearson r reported in the appendix for sensitivity. Significance threshold $p < 0.05$ unless stated.

Compute and carbon. Total compute: **[PENDING: total_a40_hrs]** A40-hours. Carbon estimate: $\text{kgCO}_2\text{e} = \text{A40-hrs} \times 0.300 \text{ kW} \times \text{grid intensity}$, using the emissions factor of the specific grid region of the compute provider at run-time as reported by the provider’s carbon dashboard. Estimated total: **[PENDING: carbon_kgco2e]** kgCO_2e .